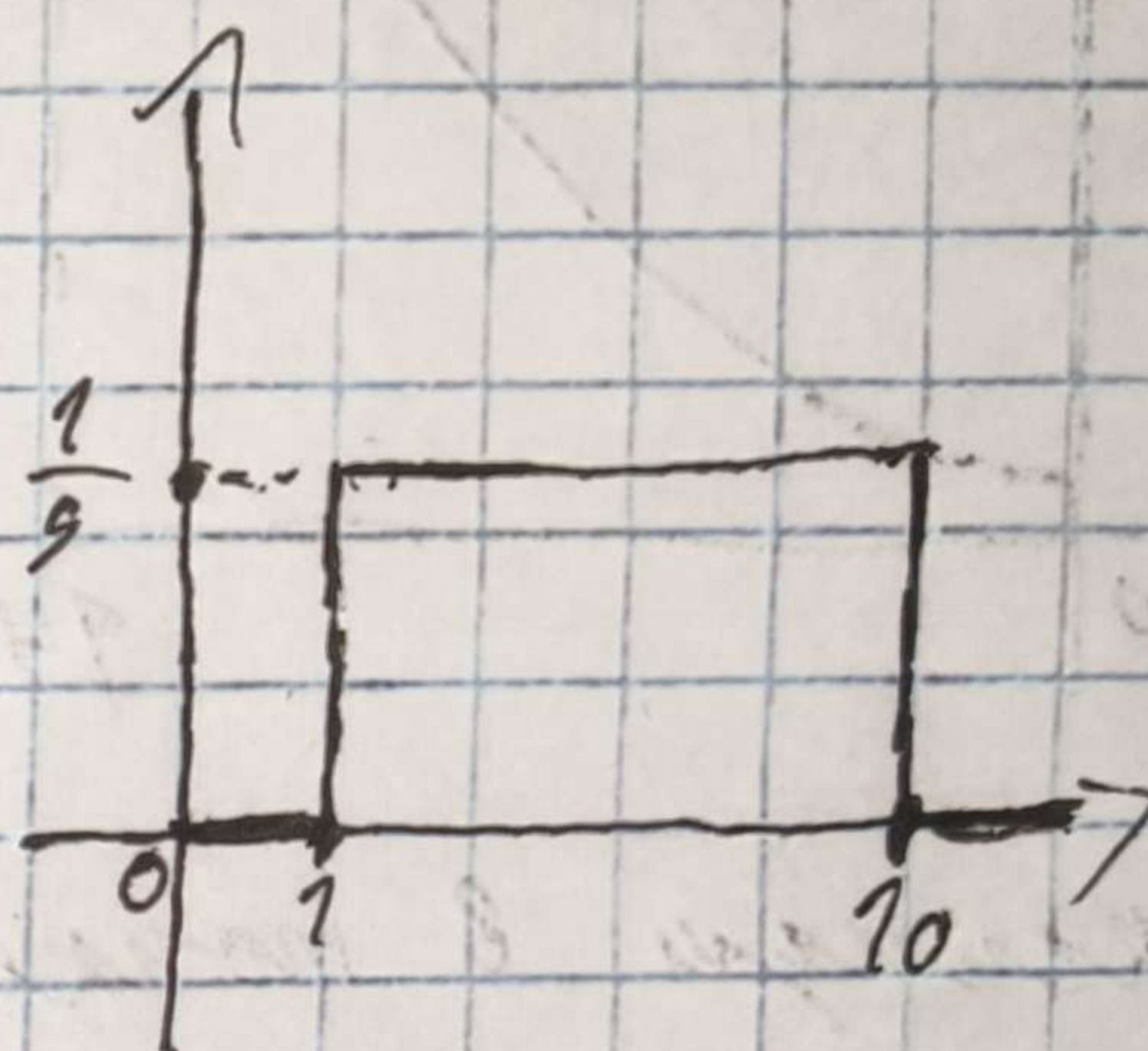


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122 равномерный, $a=1$, $b=10$, $d=8$, $\beta=13$

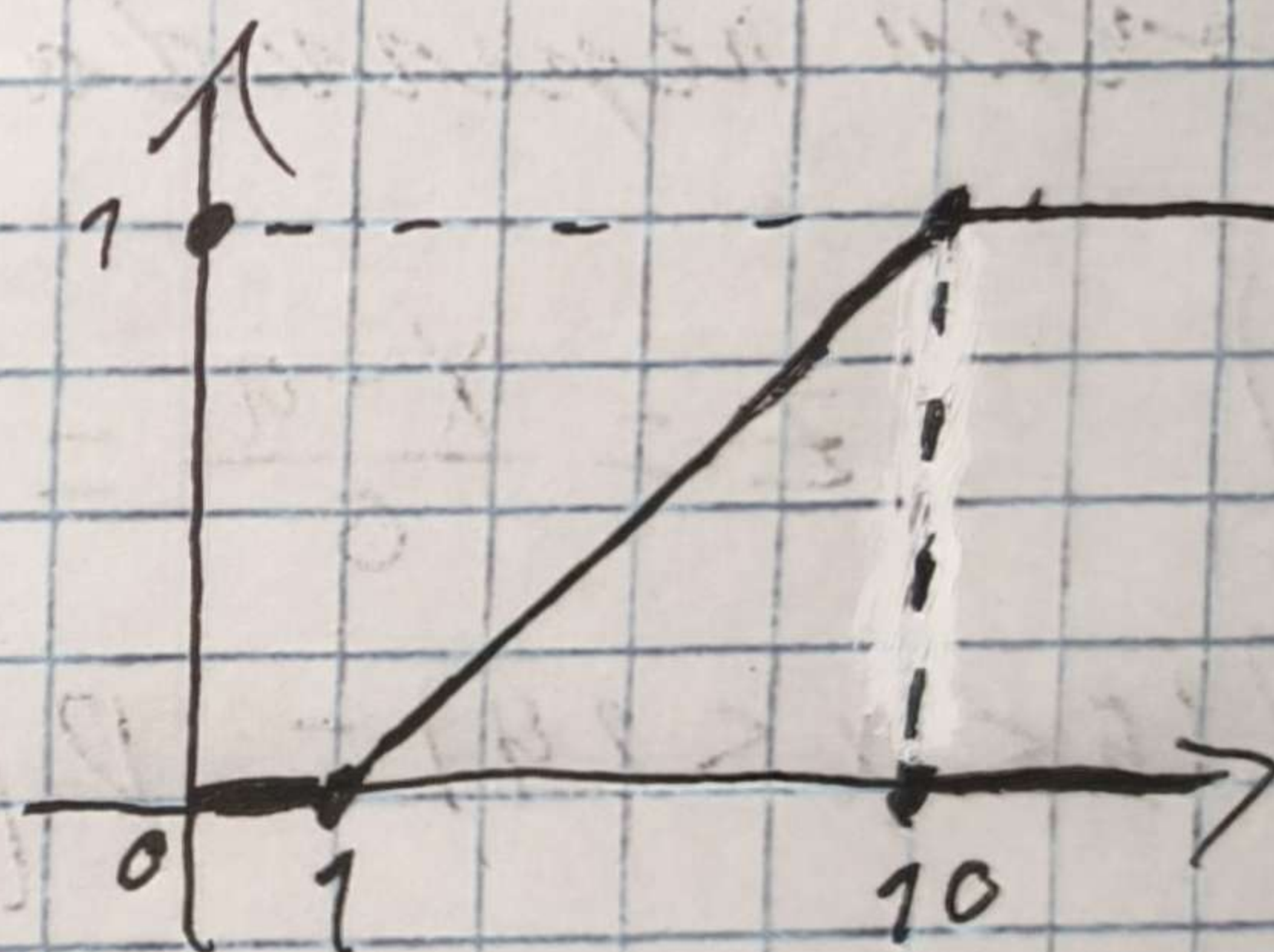
1) функция плотности распределения

$$f(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{10-1} = \frac{1}{9} & 1 \leq x \leq 10 \\ 0 & x > 10 \end{cases}$$



2) функция распределения

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{x-1}{9} & 1 \leq x \leq 10 \\ 1 & x > 10 \end{cases}$$



3)

$$Mx = \frac{a+b}{2} = \frac{1+10}{2} = 5,5$$

$$Dx = \frac{(a+b)^2}{12} = \frac{(1+10)^2}{12} = 6,75$$

$$\sigma = \sqrt{6,75} \approx 2,598$$

4)

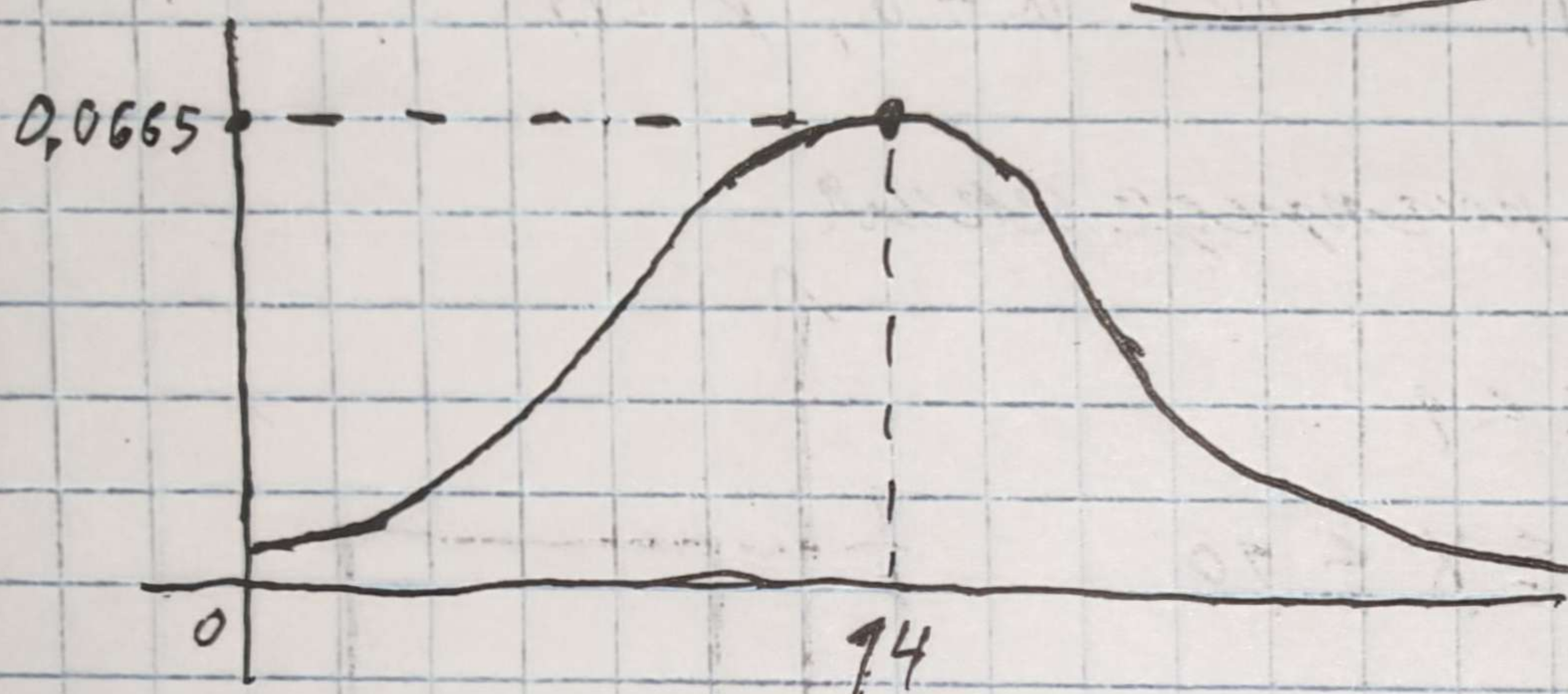
$$P(8 < x < 13) = 1 - \frac{8-1}{9} = \frac{2}{9} \approx 0,2222$$

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222 $\alpha = 11$ $\sigma = 6$ $L = 5$ $\beta = 24$ $\delta = 11$

1) $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\alpha)^2}{2\sigma^2}} = \frac{1}{6\sqrt{2\pi}} e^{-\frac{(x-14)^2}{72}}$



максимум в точке $x = 14$ $\frac{1}{6\sqrt{2\pi}} \approx 0,0665$

точки перегиба $x_1 = 14 - 6 = 8$ $x_2 = 14 + 6 = 20$

2) $Z = \frac{x-\alpha}{\sigma} = \frac{x-14}{6}$

$P(5 < x < 24) = P\left(\frac{5-14}{6} < Z < \frac{24-14}{6}\right) = \Phi(1,67) - \Phi(-1,5) =$
 $\approx 0,4522 + 0,4332 = \underline{0,8854}$

3) $P(|x-\alpha| < \delta) = 2\Phi\left(\frac{\delta}{\sigma}\right)$

$P(|x-14| < 11) = 2\Phi\left(\frac{11}{6}\right) = 2\Phi(1,83) \approx \underline{0,9328}$

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n_{ij}	Y					
	-6	0	6	12	18	24
X 1	8	1				
2	1	4	9			
3		1	10	3	4	
4			4	16	7	
5				2	13	8

$$N = 115$$

$$h_i (9, 14, 42, 27, 23)$$

$$h_j (9, 6, 23, 49, 20, 8)$$

$$\bar{x} = \frac{\sum x_i h_i}{N} = \frac{386}{115} \approx 3,357$$

$$\frac{\sum x_i^2 h_i}{N} = \frac{1450}{115} \approx 12,609$$

$$\bar{y} = \frac{\sum y_j h_j}{N} = \frac{1224}{115} \approx 10,643$$

$$\frac{\sum y_j^2 h_j}{N} = \frac{19296}{115} \approx 167,791$$

$$S_x^2 = 12,609 - (3,357)^2 \approx 1,34$$

$$S_y^2 = 167,791 - (10,643)^2 \approx 54,52$$

$$S_x \approx 1,157$$

$$S_y \approx 7,38$$

$$\frac{\sum x_i y_j h_{ij}}{N} = \frac{4962}{115} \approx 43,148$$

$$S_{xy} = 43,148 - \bar{x} \bar{y} \approx 7,42$$

$$r_{xy} = \frac{S_{xy}}{S_x S_y} \approx \frac{7,42}{1,157 \cdot 7,38} \approx 0,869$$

$$b_{yx} = \frac{S_{xy}}{S_x^2} \approx \frac{7,42}{1,34} \approx 5,537$$

$$b_{xy} = \frac{S_{yx}}{S_y^2} \approx \frac{7,42}{54,52} \approx 0,136$$

Регрессия X на Y

$$X - 3,357 = 0,136 (Y - 20,643)$$

$$X \approx 1,907 + 0,136 Y$$

Регрессия Y на X

$$Y - 20,643 = 5,537 (X - 3,357)$$

$$Y \approx -7,945 + 5,537 X$$

1

$$P = \begin{pmatrix} \frac{3}{4} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$$P_{11}^2 = \frac{3}{4} \cdot \frac{3}{4} + \frac{1}{8} \cdot \frac{1}{2} + \frac{1}{8} \cdot 0 = \frac{20}{26} = \frac{5}{8}$$

$$P_{22}^2 = \frac{3}{4} \cdot \frac{1}{8} + \frac{1}{8} \cdot \frac{1}{3} + \frac{1}{8} \cdot \frac{1}{2} = \frac{19}{96}$$

$$P_{13}^2 = \frac{3}{4} \cdot \frac{1}{8} + \frac{1}{8} \cdot \frac{1}{6} + \frac{1}{8} \cdot \frac{1}{2} = \frac{17}{96}$$

$$P_{21}^2 = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{6} \cdot 0 = \frac{13}{24}$$

$$P_{31}^2 = 0 + \frac{1}{2} \cdot \frac{1}{2} + 0 = \frac{1}{4}$$

$$P_{22}^2 = \frac{1}{2} \cdot \frac{1}{8} + \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{1}{2} = \frac{37}{144}$$

$$P_{32}^2 = 0 + \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{2} = \frac{5}{12}$$

$$P_{23}^2 = \frac{1}{2} \cdot \frac{1}{8} + \frac{1}{3} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{2} = \frac{29}{144}$$

$$P_{33}^2 = 0 + \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{3}$$

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$$P^2 = \begin{pmatrix} \frac{5}{2} & \frac{19}{96} & \frac{17}{96} \\ \frac{23}{24} & \frac{37}{144} & \frac{29}{144} \\ \frac{1}{4} & \frac{5}{12} & \frac{1}{3} \end{pmatrix}$$

$$P^3 = \begin{pmatrix} \frac{107}{192} & \frac{67}{288} & \frac{775}{576} \\ \frac{77}{244} & \frac{439}{1728} & \frac{365}{1728} \\ \frac{19}{48} & \frac{97}{288} & \frac{77}{288} \end{pmatrix}$$

$$P^4 = \begin{pmatrix} \frac{1249}{2304} & \frac{3433}{13824} & \frac{2897}{13824} \\ \frac{1825}{3456} & \frac{1333}{5184} & \frac{2227}{10368} \\ \frac{67}{244} & \frac{1021}{3456} & \frac{827}{3456} \end{pmatrix}$$

$$P^5 = \begin{pmatrix} \frac{12487805}{23887872} & \frac{74642737}{286654464} & \frac{62230673}{286654464} \\ \frac{37407757}{71663676} & \frac{112102453}{429981696} & \frac{93432701}{429981696} \\ \frac{6205729}{11943936} & \frac{9395285}{35837808} & \frac{977417}{4478976} \end{pmatrix}$$

$$\begin{cases} \frac{3}{4} \pi_1 + \frac{1}{2} \pi_2 = \pi_1 \\ \frac{1}{8} \pi_1 + \frac{1}{3} \pi_2 + \frac{1}{2} \pi_3 = \pi_2 \\ \frac{1}{8} \pi_1 + \frac{1}{6} \pi_2 + \frac{1}{3} \pi_3 = \pi_3 \end{cases} \Rightarrow$$

$$\begin{cases} \pi_1 = \frac{72}{23} \approx 0,5217 \\ \pi_2 = \frac{6}{23} \approx 0,2609 \\ \pi_3 = \frac{5}{23} \approx 0,2174 \end{cases}$$

$$\pi^* \approx [0,5217; 0,2609; 0,2174]$$

$$\pi^s \approx [0,5225; 0,2604; 0,2171]$$

Разница $\approx 10^{-3}$